Labour Supply 3: Labour Supply Over the Life Cycle

EC306- Wages and Employment

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Goals of This Section

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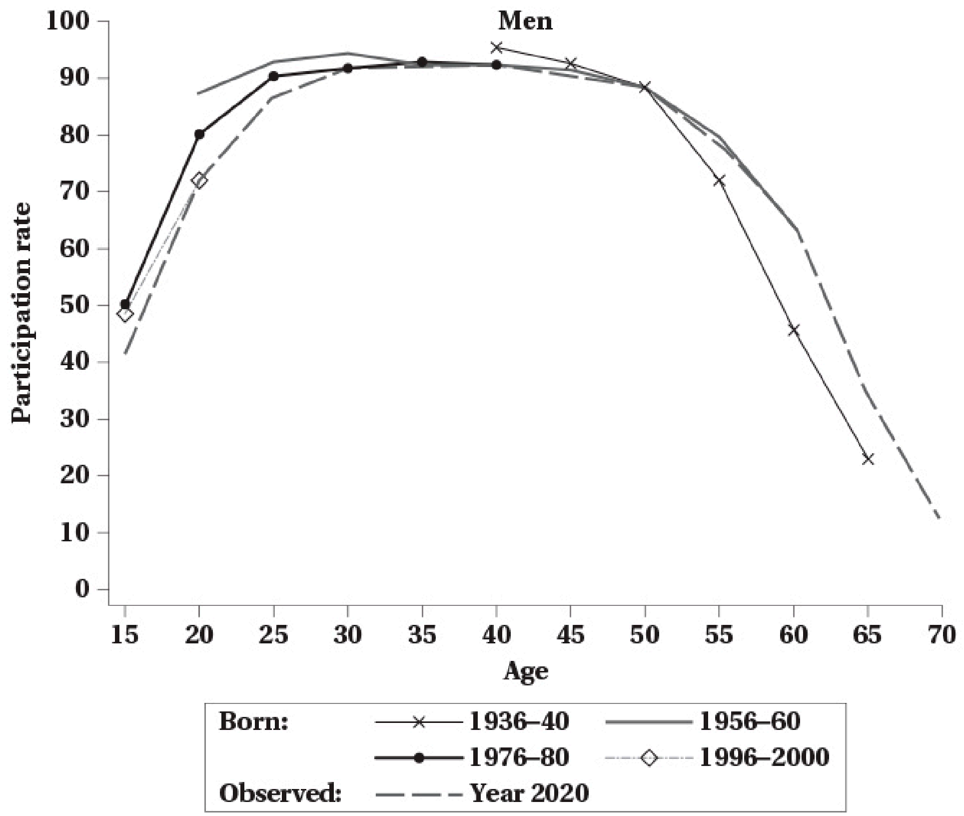
- Show patterns of labour supply over a person's lifetime
- Discuss issues comparing labour supply of people at different ages
- Introduce lifetime labour supply model
- Discuss household production
- Analyze retirement decisions

Life Cycle Labour Supply

Introduction

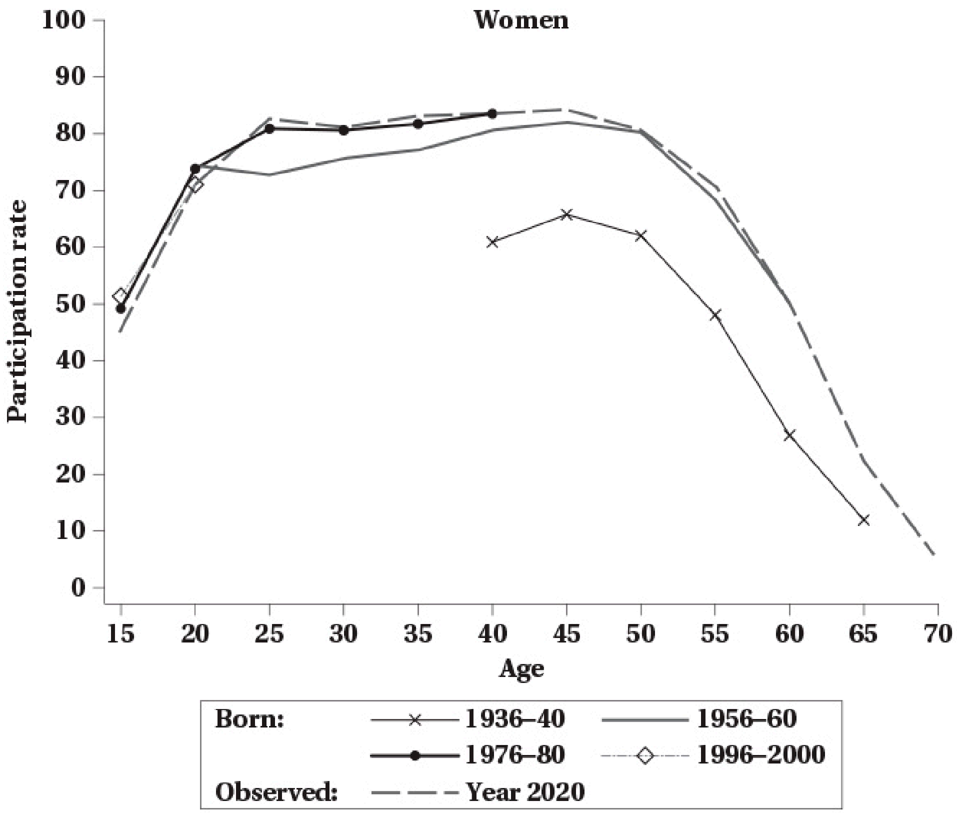
- Previous discussion focused on labour supply at a moment in time
  - Labour supply changed immediately in response to changes in wages, non-labour income, or policies
- In reality, people can plan work over their lifetime
  - Participation is highest in prime working years
  - Life events alter labour supply decisions now and in the future
    - Marriage, children, education, health, retirement
- A life cycle approach helps us understand these patterns

Life Cycle Participation for Men



- Graph follows labour supply of men born in different years
- Participation rises quickly in early 20s, peaks in late 40s, then declines
- There are differences for different birth cohorts
  - At younger ages, younger cohorts participate a bit less
  - At older ages, younger cohorts participate more

Life Cycle Participation for Women



- Women follow similar lifetime pattern
- Rates are shifted down a bit relative to men
- Participation higher for younger cohorts at all ages
    - Consistent with rising female labour force participation over time

Cohort Effects vs Age Effects

- Previous graphs highlight a technical point
- Each graph contains a line that looks at life cycle pattern in year 2020
  - Participation of people in 2020 at different ages
- Problem: people at different ages are born at different times
  - 20 year olds in 2020 born in 2000
  - 60 year olds in 2020 born in 1960
- So when we look at participation by age in a single year, we are mixing age effects and cohort effects
  - Age effects: how labour supply changes as people get older
  - Cohort effects: how labour supply differs for people born at different times

Cohort Effects vs Age Effects

- Cohort effects can be significant
  - In particular for women where it has been increasing for younger cohorts
- It is not a good approach to look at participation by age in a single year and interpret as pure age effects
- If pure age effects are of interest, better to look at participation by age for a single birth cohort
  - This is done in the graphs on the previous two slides
- Notice that the lines shift up and down for different birth cohorts
- Pattern across age is similar but there are some differences

Cohort Effects vs Age Effects

- Practically, it is difficult to do this
- Very few datasets have data on the same people over their entire life
  - Hard to track people over time
  - People die, move, refuse to respond
  - Data that does track people are often restricted access
- Another approach is to use **quasi-cohorts**
  - Group people by birth year in cross-sectional (single year) data
  - Look at how participation changes for these groups over time
  - This is what the graphs on the previous two slides do

Cohort Effects vs Age Effects

- You can do this with the Canada Census
  - Census is collected every 5 years
- Each Census collects information on age and labour force status (among many other things)
- Can follow a birth cohort across different Census years as they get older

		Census	Year		
Birth Cohort	1981	1986	1991	1996	2001
1950	31	36	41	46	51
1951	30	35	40	45	50
1952	29	34	39	44	49
1953	28	33	38	43	48
1954	27	32	37	42	47
1955	26	31	36	41	46

Cohort Effects vs Age Effects

- One caveat is that in each Census year, people in a birth cohort are different people
  - Not following the exact same individuals over time
- This is generally accepted as a reasonable approximation

Cohort Effects vs Age Effects

Born in 1975			
Observed in	2000	2005	2010
Age	25	30	35
Participation Rate	75%	80%	85%
Born in 1980			
Observed in	2000	2005	2010
Age	20	25	30
Participation Rate	75%	80%	85%

- To illustrate problem with following ages in single Census year, examine this table
- Tracks birth cohorts across Census years
- Shows age and participation rate
- For each cohort participation rises 5 points per year
- But looking at ages in a single Census year, they appear not to change

Cohort Effects vs Age Effects

- Where do cohort effects come from? Why do we expect cohorts to be different?
- People in part are defined by their generation
  - Social attitudes change over time
  - Economic conditions and policies change over time
  - Generations experience different significant events (e.g. wars, disease)
- We therefore expect cohorts to differ in their attitudes and behaviours
  - Reserach has shown women born in early 1930s had higher labour supply than previous generations
  - They were influenced by their mothers working during war
  - Male children also more likely to be married to working women later in life
- Important to separate these from effects of pure aging

Dynamic Life Cycle Models

Introduction

- We now know that labour supply varies over the life cycle
- The allocation of work over time can be modeled
- Models examine the effects on labour supply of wage changes
  - Wages can change at different ages
  - They can change for a short time or permanently
  - They can be anticipated or unanticipated
- Different types of changes have different effects on labour supply

Model

- Individuals will still maximize utility over consumption and leisure subject to a budget constraint
- The utility is a function of consumption and leisure over the lifetime

$$u = U(C_1, C_2, \ldots, C_N, L_1, L_2, \ldots, L_N)$$

- Here $U(.)$ is a function
  - e.g. could be logarithmic, cobb-douglas, etc.
- To define the budget constraint, recognize that we can consume and work in each period
- In theory, we can consume more than our income in a period by borrowing
  - We can also save some income to consume later
- We therefore need to be conscious of the interest rate on saving and borrowing

Model

- Consider a two period model
- The budget constraint says that the present value of consumption equals the present value of income

$$C_1 + \frac{C_2}{1+r} = W_1 H_1 + \frac{W_2 H_2}{1+r}$$

- r is the interest rate for borrowing and saving
- This is the present (period 1) value formulation of the lifetime budget constraint
- You can also express it in future (period 2) value

$$(1+r)C_1 + C_2 = (1+r)W_1 H_1 + W_2 H_2$$

Model

- The ability to borrow and save means a person can consume more than their income in a period
- For example, rearranging the future value of the budget we get

$$C_2 = (1 + r)(W_1 H_1 - C1) + W_2 H_2$$

- If $W_1 H_1 > C_1$, then the person saved some income in period 1 to consume in period 2
  - That savings grows with interest, and they have $(1 + r)(W_1 H_1 - C1)$ to spend in period 2
- If $W_1 H_1 < C_1$, then the person borrowed to consume more than their income in period 1
  - $(1 + r)(W_1 H_1 - C1)$ is negative and it reduces their income in period 2

Model

- In our model, the choice is between consumption and leisure
- To reexpress this in terms of leisure, note that $H_t = T - L_t$, so

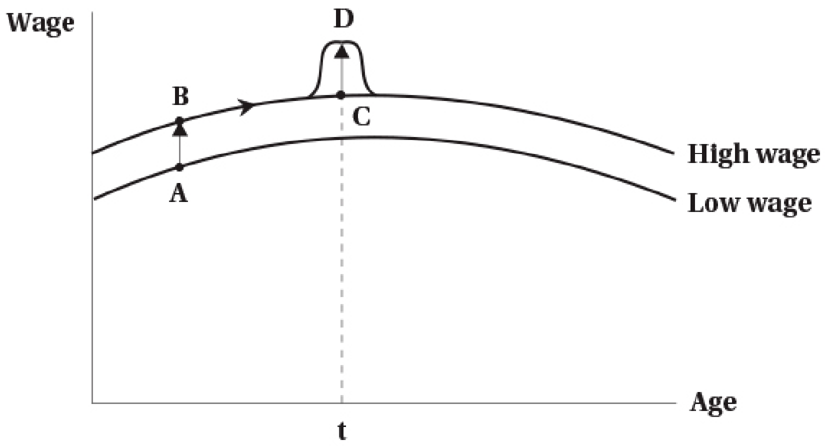
$$C_1 + \frac{C_2}{1+r} + W_1 L_1 + \frac{W_2 L_2}{1+r} = W_1 T + \frac{W_2 T}{1+r}$$

- You can consume two goods (consumption and leisure)
- The price of consumption is 1 in both periods
- The price of leisure is W_1 in period 1 and W_2 in period 2
- You consume these out of "full income"
  - Full income is the income you would have if you worked all available hours in both periods
- Period 2 values are discounted by the interest rate

Model

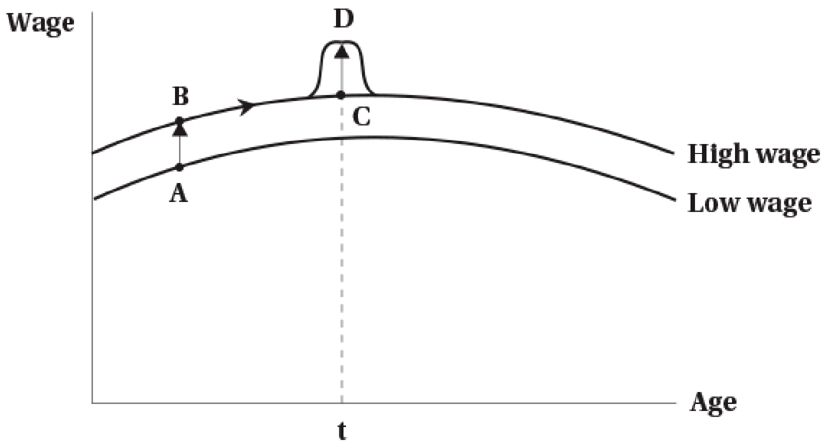
- If you maximize utility subject to the budget constraints, you get a labour supply function
  - Tells you optimal labour supply as a function of wages, interest rates, lifetime resources
- Like previous models, we can examine effects of wage changes
- There are three types
  - A permanent unanticipated change in the wage in all periods
  - An anticipated future change in the wage
  - A temporary unanticipated change in the wage in one period only
- These lead to substitution and income effects
  - More complicated because of the extra time dimension of the model
- We can see these effects in graph form

Permanent Wage Change



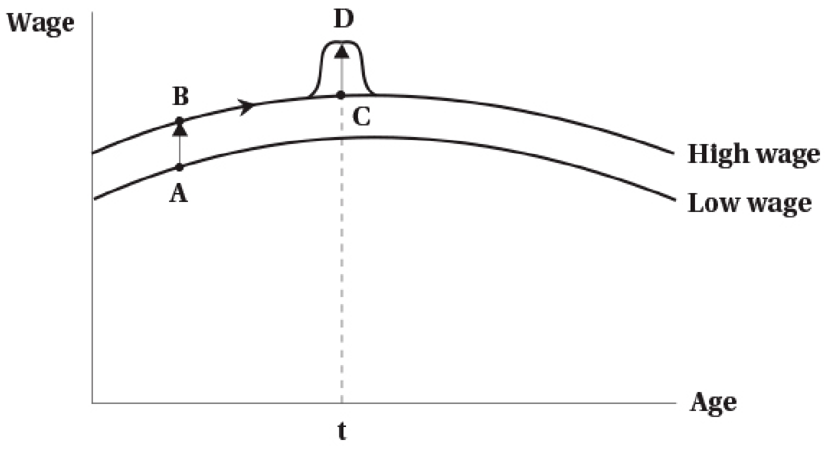
- Consider first a permanent, unanticipated change in the wage
  - Increases wage in all periods
  - Shifts wage schedule up
- Higher wage has two effects
  - Income effect: higher wage means higher lifetime income, so person consumes more leisure (less work)
  - Substitution effect: higher wage means leisure is more expensive, so person consumes less leisure (more work)
- Labour supply effect is ambiguous

Anticipated Wage Change



- An anticipated wage change moves person along their wage schedule
- There is no income effect in this case
  - Person optimized over their lifetime income
  - Lifetime income has not changed
- The substitution effect shifts leisure towards periods where wage is lower
  - If wage is higher in period 2, leisure is more expensive in period 2
  - Less leisure in period 2 and more in period 1
  - Person works more in period 2 and less in period 1

Temporary Unanticipated Wage Change



- A temporary unanticipated wage change alters wage for one period (or a small number of them) only
  - e.g. bonus pay, overtime pay
- Creates a temporary blip upward in wage schedule
- Two effects
  - Small Income effect: slightly higher lifetime income, so person consumes more leisure (less work)
  - Substitution effect: leisure is more expensive in the period with the higher wage, so person consumes less leisure (more work)

Comparing Effects of Wage Changes

- Strongest effect on labour supply is the anticipated wage change
  - Pure substitution effect
  - No offsetting income effect
- Next strongest is the temporary unanticipated wage change
  - Substitution effect
  - Small income effect
  - Person works more in period with higher wage
- Permanent unanticipated wage change has ambiguous effect
  - Substitution effect
  - Larger income effect
  - Ambiguous effect on labour supply

Evidence on Life Cycle Labour Supply

- Interesting experiment on bike messengers in Zurich
- Messengers paid a percentage of daily revenues
- Can choose their own total work hours
- Researchers randomly assigned messengers to
  - 25% higher wage for one month in September of one year (group A)
  - Others to 25% higher wage for one month in November of the same year (group B)
- Each acted as control group for the other
- Both get same increase in lifetime income, so difference is substitution effect
- Showed messengers shift labour supply to periods with higher wage

Household Production

Introduction

- So far we assumed that people choose between consumption and leisure
  - Leisure is time not working for pay
  - Encompasses pure leisure, household production, and other non-market activities
- In reality, people also spend time on household production
  - Cooking, cleaning, childcare, home maintenance, shopping, etc.
- Household production produces goods and services that can be consumed
  - Meals, clean clothes, child supervision, etc.
- In this section we explicitly model household production

Model

- Assume that people gain utility from the consumption of goods and services

$$u = f(Z_1, Z_2, \ldots, Z_N)$$

- In this model
  - $Z_1, Z_2, \ldots, Z_n$ are final goods and services
  - $f(.)$ is a function (e.g. cobb-douglas, logarithmic, etc.)
- Each of these goods and services are produced by the household using
  - A set of market inputs $X_1, X_2, \ldots, X_n$
  - Time h_i
- In this setup, time is divided between
  - Market work
  - Household production tasks

Model

- What does it mean that households “produce” the goods they consume?
- Does not mean that literally everyone is producing goods from raw materials
- Instead means that every good/service consumed requires some time to prepare
  - Hiring contractors to remodel a home requires a small amount of home production
    - Time spent explaining what you need, keeping contractor on schedule
  - Remodelling your basement by yourself requires much more time
- In this model we explicitly account for substituting time between various uses
  - Work, home production of various goods, leisure

Model

- Example: a household consumes two goods, nutrition and rest
  - These would be Z_1 and Z_2 in the utility function
- Nutrition is produced using processed or raw food, and time
  - These are X_1, X_2 and h
  - Raw food takes time to prepare equal to $h = d$ hours
- Each of the three purchased goods has a price
- Processed food
  - Price per unit of nutrition is P_c
  - Takes no time to prepare

Model

- Raw food
  - Price per unit of nutrition is $P_r + W_d$
  - The added time cost is valued at the wage rate W times d hours
- Leisure
  - Price is the wage rate W

Comparative Statics

- What happens when we start changing some of the prices?
- Increase in wage rate W
  - Cost of time affects both rest and production of raw food
  - Higher wage makes home production more expensive
    - Time spent cooking becomes more expensive
    - Substitute away from home production to processed food
    - Substitution effect: less home production, more processed food
    - Income effect: higher wage means higher income, so more consumption of all goods
  - Effect on rest is same as before
    - Substitution effect: less rest, more work
    - Income effect: higher income, so more rest

Comparative Statics

- Decrease in price of processed food P_c
  - Processed food becomes cheaper relative to other goods
  - Substitution effect: more processed food, less home production
  - Income effect: lower price of processed food means higher real income, so more consumption of all goods
- Decrease in price of raw food P_r
  - Raw food becomes relatively cheaper
  - Substitution effect: more raw food, less processed food
  - Income effect: lower price of raw food means higher real income, so more consumption of all goods

Comparative Statics

- Decrease in time required to prepare raw food d
  - Example: technology that reduces cooking time (e.g. Instant Pot)
  - Home production becomes relatively cheaper (through lower time cost)
  - Substitution effect: more raw food, less processed food
  - Additional substitution effect: higher real wage (i.e. they keep more of their wage), less rest, more work
  - Income effect: lower time cost of raw food means higher real income, so more consumption of all goods

Discussion

- Model adds more real world complexity to match what people actually do
- Emphaszies that people do not just choose between work and leisure
- Also emphasizes the role of the household relative to the individual
  - Household members can specialize in different tasks
  - Changes for one household member can affect others
- Discussion above did not mention household production function, but it is important
  - Different households have different abilities to produce goods
  - This can affect their choices and responses to changes in prices

Fertility and Childbearing

Introduction

- Fertility is a key life event that affects labour supply
  - Particularly for women, but also men to some degree
- Children are typically planned to some degree
  - Timing of children is often chosen to fit with work plans
  - Number of children is often chosen based on lifetime work plans
- Economists have developed models to analyze fertility decisions
  - These models are similar to the life cycle labour supply model
- We will not go over the model because it is a bit beyond the scope of this course
- But we will discuss some of the key variables that affect fertility decisions

Variables Affecting Fertility - Income

- Income is positively related to the number of children
  - Higher income means higher ability to pay for children
  - Children are a “normal good”
- Some disagreement on whether children are considered a normal good in our models
  - Empirically, higher income families and countries have lower fertility rates
  - Explanation is that substitution effect of higher income dominates income effect
    - Higher income means higher opportunity cost of time spent on children

Variables Affecting Fertility - Price of Children

- Like goods, the demand curve for children slopes down
  - Higher “price” of children means fewer children are chosen
- Several elements involves in the price of children
  - Direct costs: food, clothing, housework, etc.
  - Opportunity costs: time spent caring for children could be spent working for pay
- A higher price (wage rate) creates income and substitution effects
  - Income effect: higher price means higher lifetime income, so more children
  - Substitution effect: higher price means time with children is more expensive, so fewer children
- Effect is theoretically ambiguous, but empirically substitution effect dominates

Variables Affecting Fertility - Price of Related Goods

- Many goods are consumed as part of having children
  - Childcare, education, etc.
- These are complementary goods
  - If you have more children, you will consume more of these goods
  - You could also consider these direct costs of children
- We know that an increase in the price of a complementary good reduces demand
  - Higher price of childcare or education means fewer children

Variables Affecting Fertility - Tastes

- Tastes for children vary across people and over time
  - Some people really want children, others do not
  - Some cultures value children more than others
  - Attitudes towards children have changed over time
- A stronger taste for children means more children
  - This is like a shift in the demand curve for children
- Over time in Canada, there has been a shift away from children
  - Fertility rates have declined over time
  - People are having fewer children than in the past

Variables Affecting Fertility - Technology

- Technology in this case could mean several things
  - Could refer to the “production” of children
  - Might also relate to the goods and services used with children
    - Home automation, disposable diapers, baby monitors, carriers
- Some key developments in the last few decades
  - Birth control: allows people to better plan timing and number of children
  - Fertility treatments: allows some people to have children who otherwise could not
  - Medical care: reduces infant mortality, so fewer children needed to ensure some survive
  - Home technology: reduces time cost of childcare, so lower opportunity cost of children
- Higher technology in any of these areas tends to increase demand for children

Empirical Evidence

- Economists have examined the relationship between income and fertility
- Difficult because of omitted variables
  - A simple correlation between income and fertility tends to be negative
  - Important to hold constant other variables that affect both income and fertility
    - Example: education levels are positively related to income and negatively related to fertility
- When other variables are held constant, income tends to be positively related to fertility
  - Consistent with children being a normal good
- Tax incentives also tend to increase fertility

Retirement and Pensions

Introduction

- Retirement is a key life event that affects labour supply
  - Most people retire at some point
  - Retirement age has been changing over time
- It can mean many things
  - Permanent exit from the labour force
  - A reduction to part-time work
  - Retiring from one full-time job and taking another
- For most people retirement means permanent exit from the labour force

Determinants of Retirement - Age

- Age is the most important determinant of retirement
  - Most people retire between ages 60 and 70
  - Very few people retire before age 50 or after age 75
- Some places have mandatory retirement
  - In Canada it is generally illegal to force retirement based on age
  - Except if deemed a bona fide occupational requirement (BFOR), like for pilots, police, firefighters, air traffic controllers
  - Still exits in other parts of the world (e.g. Japan, China, parts of Europe)

Determinants of Retirement - Wealth/Earnings

- Wealth in our model entered as non-labour earnings
  - Higher wealth means higher non-labour income
  - Higher non-labour income means more leisure (less work)
- Earnings changes have income and substitution effects
  - Income effect: higher earnings means higher lifetime income, so more leisure (less work)
  - Substitution effect: higher earnings means leisure is more expensive, so less leisure (more work)
- Can also think of this in a lifecycle. model

Determinants of Retirement - Other

- Health is an important determinant of retirement
  - Poor health can force people to retire early
  - Good health can allow people to work longer
  - Health of family members can also play a role
  - Might operate through non-labour income or preferences
- Changing nature of work can influence decisions
  - Move towards less physically demanding jobs can prolong work life
  - More flexible work arrangements can allow people to work longer
  - Alternatively, more stressful jobs can lead to earlier retirement
  - Automation might force early retirement
  - Could affect the budget constraint or preferences

Determinants of Retirement - Other

- Changing nature of family
  - There are many more two-earner families now
  - This can lead to earlier retirement for one spouse
  - Or could prolong work if spouses want to work together longer
  - Children living at home for longer can also affect retirement timing
  - Increase in caring for elderly parents can lead to earlier or later retirement
  - Might affect budget constraint or preferences

Retirement Income

- Public and private pension plans have major effect on retirement decisions
  - They determine most income in retirement
- The pension system in Canada consists of

1. Old Age Security (OAS)

- A flat rate pension available to most Canadians over age 65
- Funded out of general tax revenues
- Monthly max is $740 for those aged 65-74, $814 for those 75+
- Clawback for high income earners (above $148,000 in 2024)
- Supplemented by the Guaranteed Income Supplement (GIS) for low income seniors

Retirement Income

2. Canada/Quebec Pension Plan (CPP/QPP)
    - Earnings-related pension available to those who have contributed
    - Funded by contributions from employees, employers, and self-employed
    - Monthly max is $1,433 in 2025 (age 65)
    - Can start receiving as early as age 60 (reduced) or as late as age 70 (increased)
    - Based on average earnings over working life, up to a maximum limit
3. Employer Pension Plans
    - Many employers offer pension plans to employees
    - Can be defined benefit or defined contribution plans
    - Vary widely in terms of benefits and eligibility
    - Includes private savings from RRSPs

Old Age Security

- What effect does OAS have on retirement decisions?
- Operates as a lump sum transfer
  - Increases non-labour income
  - Creates income effect: more leisure (less work)
  - No substitution effect because it does not change the wage rate
- GIS operates like a negative income tax
  - People get a the lump sum payment
  - Payments reduced by $0.50 for each dollar of work income
  - Has work incentives equal to a negative income tax

Canada/Quebec Pension Plan

- The effect of pension income on retirement depends on how it is structured
- In Canada currently, it is simply a fixed payment
  - Increases non-labour income
  - Creates income effect: more leisure (less work)
  - No substitution effect because it does not change the wage rate
- In the past, there was a provision that reduced benefits if you worked
  - Similar to a negative income tax
  - If the tax back rate is 100%, then it is like pure welfare
  - Could also not tax back until a certain income level

Employer Pension Plan

- Generally involve a fixed payment based on years of service and earnings
  - Increases non-labour income
  - Creates income effect: more leisure (less work)
  - No substitution effect because it does not change the wage rate
- Over time the parameters that determine the fixed payment can change
  - This can alter work incentives at retirement
  - Example: if the benefit formula is changed to require more years of service
    - Would reduce the income effect of the pension
    - Could also create a substitution effect to work more to get the pension

Evidence on Effect of Pensions

- Studies find that pensions affect retirement in expected ways
  - Pensions that increase non-labour income lead to earlier retirement
  - Pensions that create work incentives lead to later retirement
- More recent work
  - Increase in retirement age increases working-age disability claims in UK
  - Increasing public pension participation improves health behaviours (less smoking)
  - Pension reform allowing Japanese spouses to claim half of pension earnings increased divorce

Summary

Summary

- Labour supply varies over the life cycle
  - Participation highest in prime working years
- A life cycle approach helps us understand these patterns
- Dynamic life cycle models can analyze effects of wage changes
  - Permanent unanticipated changes
  - Anticipated future changes
  - Temporary unanticipated changes
- Household production is an important part of labour supply decisions
  - People allocate time between work, household production, and leisure
  - Changes in prices of goods and time costs affect these choices
- Fertility decisions are affected by income, prices, tastes, and technology
- Retirement decisions are affected by age, wealth, health, family, and pensions